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The Scientific Method in *Mythbusters*: Blowing Up a Manhole!

Season 7 Episode 15 of the T.V. show *Mythbusters* is inspired by the fastest man-made object ever. In the late 1950's, a series of nuclear experiments known as Project Plumbob were conducted in Nevada. During one specific test, a nuclear warhead was buried in a hole in the ground, which was covered by a manhole cover welded shut. Upon detonation, the welds broke and the manhole cover was sent skyward at several times Earth's escape velocity. For the show, they do not mention bombs, instead using methane buildup as the source of the detonation, and the question they are trying to answer is "can a manhole fly?"

They begin by constructing a scale model of a sewer which will allow them to test the viability of their methods before doing a full-scale test of their hypothesis. They found the correct stoichiometry to cause detonation and filled their scale model with it. Their first test here was with an open ended sewer, but the deflegtation took the path of least resistance, causing the manhole covers to stay in place. Then then capped off the end of the mini sewer, which allowed the deflegtation to force the manholes out of the holes they covered. This proved their larger scale experiment may be viable. They then added debris to the system, which impedes the shock wave propagating through the pipe. This drastically increased the energy imparted onto the manhole by causing turbulence in the gas mixture, increasing the combustible surface are. This small scale experiment allowed them to properly set their experiment parameters. They used the open ended sewer as a "control" in this to show that a closed system was needed to cause the manhole covers to actually leave the ground. Other than that there wasn't much of a control due to the nature of the experiment and their production budget restrictions.

They then scaled the experiment up to a real sized sewer. They went out into the desert and had contractors come and build a to-scale sewer system using the same techniques that are used to build functional sewers. They then added chain link fencing to act as an obstruction to make the shock wave turbulent, hooked up their methane tank to a solenoid so they could safely fill the system, and buried the whole thing. This was an ideal scenario for the manholes to fly, which was chosen since there was a high likelihood they would only get one shot at the experiment. After the fire department arrived and the sewer was filled with methane, they sparked the mixture. The manhole covers shot very high into the air and even caused the ground to become softer in the area around the sewer they built.

They collected their displacement, velocity, and acceleration date using high speed cameras and specific paint patterns on the manhole covers to keep track of the center of mass of each of them. They also had at least one barometer in the sewer to record air pressure within the system. After analyzing the high speed footage and the barometric readings, they concluded that the highest manhole cover reached a height of 150 feet off of the ground and the sewer had a peak pressure of 100 pounds per square inch. This data allowed them to confirm the myth that manhole covers can fly if methane buildup in a sewer system is ignited under the correct conditions. This also indirectly confirms the "myth" of the fastest man made object in history. They proved that this worked with methane deflegtation, thus it stands to reason that this principle would hold using nuclear detonation as the source of kinetic energy, assuming the vessel could contain such a blast.